

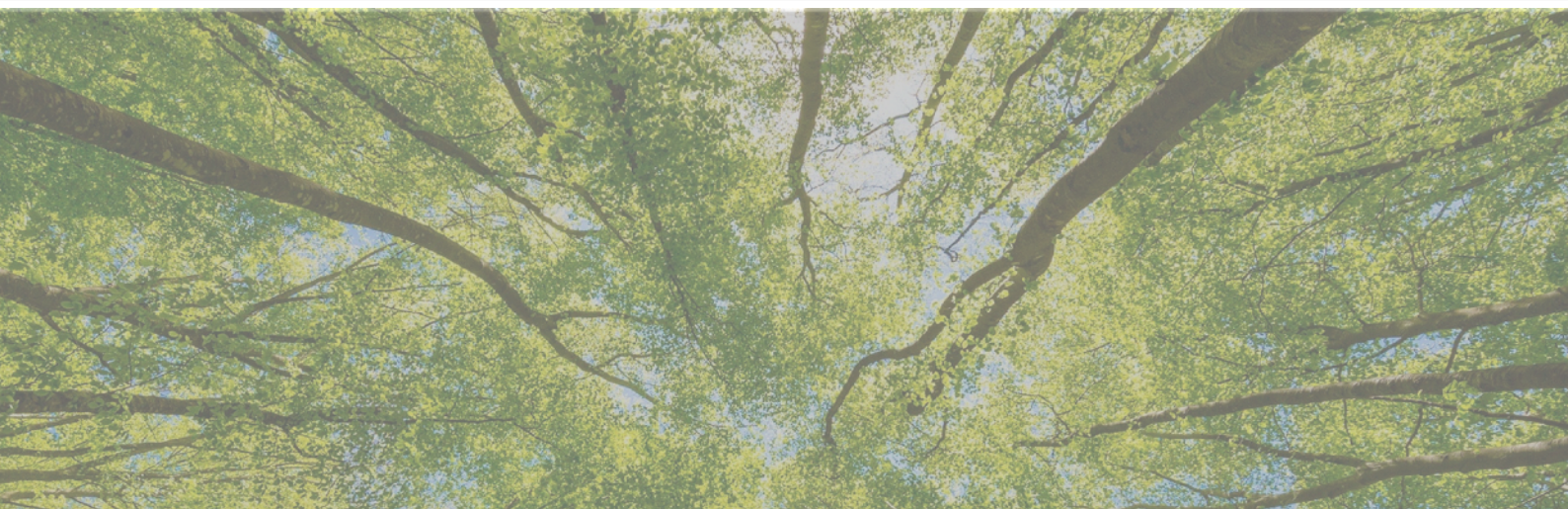


Analysis Report

Risk Factors and Living Area Impact – UrbanTrace Dashboard

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1. Introduction

This report presents an **interactive dashboard** built using the **UrbanTrace** dataset, which predicts individual pollution exposure and associated risk levels based on lifestyle and environmental factors.

The main objective is to **identify key factors contributing to increased risk** and to examine how the **risk index varies according to the type of residential area** (urban, semi-urban, rural) and other environmental and lifestyle factors.

The dataset is **synthetic**, collected from a Kaggle source, and contains 10,000 rows and 14 columns. Despite being synthetic, the dataset provides meaningful patterns that help illustrate realistic relationships between environment, lifestyle, and pollution-related risk.

2. Methodology

The analysis followed a structured workflow:

- **Data Cleaning:** Ensured consistency, removed blank spaces, standardized categorical text, created helper boolean columns, filled missing values with “Unknown”, verified target column consistency, checked for duplicate columns, and reviewed extreme values without removing them.
- **Exploratory Data Analysis (EDA):** Pivot tables and KPIs were used to summarize key patterns and relationships between environmental/lifestyle factors and risk levels.
- **Dashboard Construction:** An interactive Excel dashboard with macros was created to dynamically explore risk scores based on residential area, awareness level, years in location, and green space access.

3. Data Overview

3.1 Variables

The dataset includes the following variables:

Column	Description
daily_travel_time	Daily commute time (minutes)
vehicle_ownership	Vehicle type owned
location_type	Urban, Semi-Urban, or Rural
nearby_industries	Number of nearby factories
green_space_access	Low, Moderate, High
home_air_quality	Indoor air quality score (0–100)
work_location_type	Office, Factory, Remote, etc.
smoker_in_household	Any indoor smoker at home
noise_pollution_level	Noise exposure (0–100)
use_of_air_purifiers	Air purifier usage
awareness_level	Pollution awareness level
years_in_location	Time lived at current address (years)
pollution_exposure_score	Target: overall exposure score (0–100)
risk_category	Target: risk level (Low, Medium, High)

3.2 Data Cleaning

The raw dataset contained 10,000 rows and 14 columns, some of which included missing or inconsistent values. To ensure a reliable analysis, the following data cleaning steps were performed:

1. Missing Values Assessment:

- Columns with missing values were identified, and the percentage of missing data was calculated for each.
- Missing values were checked for distribution across key categories (e.g., urban, semi-urban, rural) to avoid bias in the analysis.
- Since missing values represented a small fraction of the dataset and were proportionally distributed, no imputation was performed.

2. Data Transformation and Standardization:

- Missing values were filled with “Unknown”.
- Additional boolean columns were created to facilitate counting of certain factors.
- The data types for each column (e.g., numeric, percentage, categorical) were explicitly defined.

3. **Variable Type Verification:**

- Numerical columns (daily_travel_time, nearby_industries, home_air_quality, noise_pollution_level, pollution_exposure_score, years_in_location) were confirmed to be numeric and within expected ranges.
- Categorical columns (vehicle_ownership, location_type, green_space_access, work_location_type, smoker_in_household, awareness_level, risk_category) were checked for valid categories.
- A check for duplicate columns was also performed to confirm that no repeated variables existed in the dataset.

4. **Outlier Detection:**

- Extreme values were reviewed to ensure they represented realistic observations and not data entry errors. No rows with extreme values were removed, as retaining them allows the analysis and dashboard to reflect the full range of possible scenarios and better illustrate differences in risk levels across environments.

5. **Outlier Detection:**

- The relationship between the two target columns (pollution_exposure_score and risk_category) was examined to ensure that the continuous risk score (0–100) was correctly classified into the three risk categories (Low, Medium, High).
- This verification confirmed that the classification logic was consistent and that all entries matched their corresponding risk category before proceeding to EDA and dashboard construction.

By applying these steps, the dataset was prepared to ensure **accuracy, consistency, and reliability**, allowing subsequent analysis to generate meaningful and interpretable insights.

4. Exploratory Data Analysis (EDA)

The exploratory analysis was conducted using pivot tables in Excel to summarize key patterns and relationships between environmental/lifestyle factors and risk levels. The following analyses were performed:

4.1 Key Performance Indicators (KPIs)

- Average pollution exposure score
- Percentage of population in the high-risk category
- Average home air quality rating
- Percentage of people without access to green spaces

These KPIs provide an overall view of the population's environmental risk and living conditions.

4.2 Factor Analysis

- Factors with the strongest impact on environmental risk were identified using pivot tables comparing risk categories against variables such as green space access, nearby industries, and awareness level.
- Relationship between years lived and living place: examined how duration at the current residence influences risk in different areas.
- Impact of living environment and use of air purifiers on risk: analyzed how these factors interact with overall risk.

4.3 Other Variables

- Work location vs living place: no significant insights were obtained.
- Average daily travel time: did not vary meaningfully across urban, semi-urban, and rural areas.
- Noise pollution vs living place: average values were similar across all residential types.

These results reflect the nature of the synthetic dataset, where some relationships are perfectly represented while others may not correspond to real-world patterns. Despite this, the analyses allowed for logical and coherent observations regarding environmental risk factors.

5. Key Results and Conclusions

The analysis of the UrbanTrace dataset revealed several important insights regarding environmental risk factors and the influence of living environment on pollution exposure:

- **Overall KPIs:** The average pollution exposure score provides a baseline for environmental risk. A notable portion of the population falls into the high-risk category, while some lack access to green spaces, highlighting potential environmental inequalities.
- **Factors Influencing Risk:** Access to green spaces, proximity to industries, and awareness level are among the factors with the strongest impact on risk. Individuals living in urban areas consistently show higher pollution exposure scores compared to semi-urban and rural residents. High-risk individuals tend to live closer to industries, have lower access to green spaces, and exhibit poorer home air quality.
- **Other Observations:** Daily commute time and noise exposure averages were similar across residential types, providing limited discriminatory value. Work location versus living place did not reveal significant patterns in this dataset.
- **Conclusions:** The synthetic nature of the dataset explains why some relationships are perfectly represented (e.g., proximity to industries vs risk) while others show little or no correlation (e.g., noise exposure vs living place). Despite these limitations, the analyses produced logical and consistent patterns, supporting the expected impact of environmental and lifestyle factors on pollution risk. The dashboard successfully illustrates how different factors interact with living environment to influence risk scores, providing a clear visual tool for exploring environmental exposure scenarios. This approach could be applied to real-world datasets to validate and expand these findings.